

Word Embedding Analysis in LLMs

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Relevant Code: [\[Colab\]](#)

1. Dataset & Model

Dataset: The *BookCorpus* was utilized, from which 2429 sentences containing either “love” or “romance” were extracted. Sentences were filtered to be between at least 5 words.

Model: A pretrained “bert-base-uncased” model was used.

2. Grouping Method

For sentence grouping based on cosine similarity, three different approaches were explored: (1) **greedy algorithm**, (2) **graph-based grouping**, (3) **DBSCAN**. All approaches are considered to avoid duplicate counts in groups.

Since the objective was to determine the number of meaningful group based on sentence similarity, DBSCAN was chosen as one of the methods due to its ability to form clusters in an unsupervised manner without pre-defining the number of clusters (unlike K-Means).

Given that the extracted sentences contained the words “love” or “romance”, the cosine similarity for the first few sentences was higher than in previous analyses. Therefore, a threshold of 0.85 was set for cosine similarity.

From the results as below, it was observed that both Graph-based clustering and DBSCAN introduced potential noise in the clustering process. Due to this, the **Greedy algorithm** was selected as the primary grouping method.

```
Greedy Clustering (Final Layer): 175 clusters
Greedy Clustering (Intermediate Layer): 120 clusters
Graph-based Clustering (Final Layer): 20 clusters
Graph-based Clustering (Intermediate Layer): 8 clusters
DBSCAN Clustering (Final Layer): 2 clusters
DBSCAN Clustering (Intermediate Layer): 2 clusters
```

```
--Greedy Grouping--
Final layers' Group Counts of first 5th group: [906, 764, 13, 4, 139]
Intermediate layers' Group Counts of first 5th group: [1185, 398, 45, 3, 152]
--Graph Grouping--
Final layers' Group Counts of first 5th group: [2410, 1, 1, 1, 1]
Intermediate layers' Group Counts of first 5th group: [2422, 1, 1, 1, 1]
--dbscan Grouping--
Final layers' Group Counts of first 5th group: [2410, 19]
Intermediate layers' Group Counts of first 5th group: [2422, 7]
```

To validate that there are no duplicates sentences within the greedy algorithm groups, a sanity check was performed:

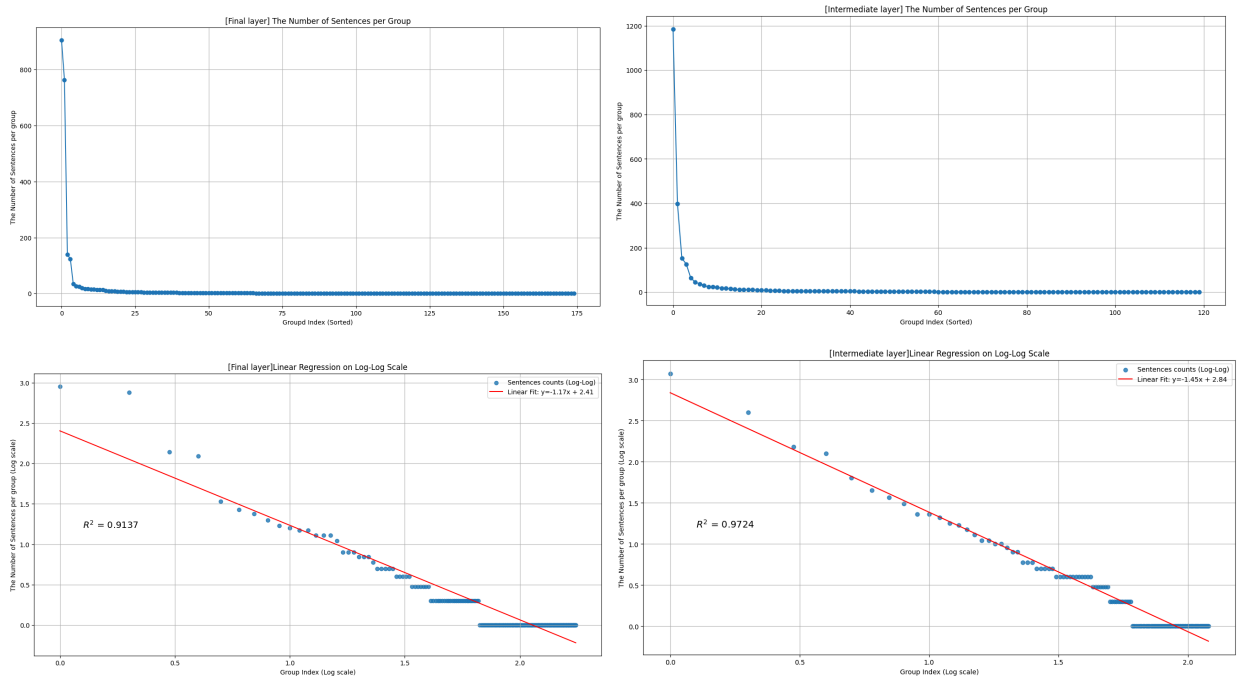
```
print("sanity check using the number of sentences in each group")
print("The total number of sentences in the Final layers:", sum(graph_group_counts_final))
print("The total number of sentences in the Intermediate layers:", sum(graph_group_counts_inter))

print("Sanity check result:", n == sum(graph_group_counts_final) == sum(graph_group_counts_inter))
```

sanity check using the number of sentences in each group
The total number of sentences in the Final layers: 2429
The total number of sentences in the Intermediate layers: 2429
Sanity check result: True

3. Result and Analysis

For the intermediate layer, the 8th layer in BERT was chosen for extracting context vectors, based on its position as the 2nd/3rd layer from the beginning. To analyze the grouping structure, the number of sentences per group was plotted for the final layer and intermediate layer embeddings.



As observed in the log-log linear regression plots, **the slopes indicate a strong structure when using the *BookCorpus* dataset** compared to previous experiments with *Wikipedia*-based continuous sentences. (For references, when using *Wikipedia*, the slope is -0.55 and R^2 is 0.9048). These results suggest that using *BookCorpus* sentences with specific words, such as “love” or “romance”, leads to a stronger correlation in sentence grouping than using *Wikipedia* continuous sentences.

4. Future work

Moving forward it may be beneficial to explore literature-specific datasets containing “love” or “romance” themes instead of relying on general-purpose datasets like *BookCorpus*. This would allow for extracting more continuous sentences within a single narrative, potentially improving the coherence of similarity-based grouping.